

NETSDK 应用指南

V1.4

目录

NETSDK 底看提博	1
修订记录	1
1.概述	2
2.约定和缩写	2
3.传输机制	2
4.设备搜索	2
5.	2
5.1 System	2
5.1.1 Device Information	2
5.1.2 System Local Time	2
5.1.3 System Time Ntp	3
5.2 Audio	3
5.2.1 Audio Input Channel List	3
5.2.2 Audio Input Channel	3
5.2.3 Audio Encode Channel List	4
5.2.4 Audio Encode Channel	4
5.3 Video	5
5.3.1 Video Input Channel List	5
5.3.2 Video Input Channel	5
5.3.3 Video Input Channel Sharpness Level	5
5.3.4 Video Input Channel brightness Level	6
5.3.5 Video Input Channel flipEnabled	6
5.3.6 Video Input Channel mirrorEnabled	6
5.3.7 Video Input Channel Saturation Level	7
5.3.8 Video Input Channel Hue Level	7
5.3.9 Video Input Channel Contrast Level	7
5.3.10 Video Input Channel Privacy Mask List	8
5.3.11 Video Input Channel Privacy Mask	8
5.3.12 Video Encode Channel List	8
5.3.13 Video Encode Channel	8
5.3.14 Video Encode Channel Name Overlay	9
5.3.15 Video Encode Datetime Overlay	9
5.3.16 Video Encode Deivce ID Overlay	10
5.3.17 Video Encode Text Overlay List	10
5.3.18 Video Encode Text Overlay	10
5.3.19 Video Encode Request Key Frame	11
5.3.20 Video Encode Snap Shot	11
5.3.21 Video Motion Detection Channel List	11
5.3.22 Video Motion Detection Channel	11
5.3.23 Video Motion Detection Channel Status	12
5.4 IO	12
5.4.1 IO Alarm Input Channel List	12

5.4.2 IO Alarm Input Channel.....	12
5.4.3 IO Alarm Input Channel Port Status.....	13
5.4.4 IO Alarm Output Channel List.....	13
5.4.5 IO Alarm Output Channel.....	13
5.4.6 IO Alarm Output Channel Port Status.....	14
5.4.8 IO Alarm Output Channel Trigger.....	14
5.5 PTZ.....	14
5.5.1 PTZ Channel List.....	14
5.5.2 PTZ Channel.....	15
5.5.3 PTZ Channel Control.....	15
5.6 Network Transport Stream.....	16
5.6.1 Network Transport Stream Channel List.....	16
5.6.2 Network Transport Stream Channel.....	16
5.7 Network.....	17
5.7.1 Network Interface List.....	17
5.7.2 Network Interface.....	17
5.7.3 Network Interface Lan.....	17
5.7.4 Network Interface Pppoe.....	18
5.7.5 Network Interface Dns.....	18
5.7.6 Network Interface Wireless.....	18
5.7.7 Network Port List.....	19
5.7.8 Network Port.....	19
5.7.9 Network Dns.....	19
5.7.10 Network Esee.....	19
5.8 SDCard.....	20
5.8.1 SDCard media search.....	20
5.8.2 SDCard media playbackFLV.....	20
5.8.3 SDCard format.....	20
5.8.4 SDCard status.....	21
5.9 Image.....	21
5.9.1 Image.....	21
5.9.2 IrCutFilter.....	21
5.9.3 ManualSharpness.....	22
5.9.4 Denoise3D.....	22
5.9.5 WDR.....	22
5.9.6 AF.....	23
Appendix I Data Structure.....	23
DeviceInfo.....	23
TimeNtp.....	24
AudioInputChannel.....	24
AudioEncodeChannel.....	24
VideoPrivacyMask.....	25
VideoInputChannel.....	25
VideoChannelNameOverlay.....	26

VideoDatetimeOverlay.....	26
VideoDeviceIDOverlay.....	27
VideoTextOverlay.....	27
VideoEncodeChannel.....	27
VideoMotionDetectionGrid.....	28
VideoMotionDetectionRegion.....	28
VideoMotionDetectionChannel.....	29
AlarmInputChannel.....	29
AlarmOutputChannel.....	29
AlarmIOPortStatus.....	30
PTZChannel.....	30
StreamTransportRTSP.....	30
StreamTransportRTSPoverHTTP.....	31
StreamTransportFLVoverHTTP.....	31
StreamTransportRTMP.....	31
StreamChannel.....	32
NetworkInterfaceList.....	32
NetworkInterfaceLan.....	33
NetworkInterfaceUpnp.....	33
NetworkInterfacePppoe.....	33
NetworkInterfaceDns.....	33
NetworkInterfaceWireless.....	34
NetworkInterfaceStationMode.....	34
NetworkInterfaceAccessPointMode.....	34
NetworkInterfaceDhcpServer.....	35
NetworkPortList.....	35
NetworkDNS.....	35
NetworkEsee.....	35
SDCardDbMedia.....	35
Image.....	36
/* refers to < ImageManualSharpness>, */.....	36
IrCutFilter.....	36
ManualSharpness.....	36
Denoise3d.....	37
WDR.....	37

修订记录

Revision History	Description	Author	Data
1.0.0 Revision 1908	第一次发布。	罗展昭	2013-07-10
1.1 Revision 2001	重新定义数据格式，增加音频输入、音频编码、视频输入、运动侦测、报警 IO 相关接口。	罗展昭	2013-08-10
1.1 Revision 2017	实现视频输入、音频输入、音频编码接口。	朱辉文、卢冠霖	2013-08-12
1.1 Revision 2001	实现视频编码分辨率修改接口。	罗展昭	2013-08-15
1.1 Revision 2108	加入时间设置接口/System/localTime，统一时间格式为 ISO8601。	卢冠霖	2013-08-16
1.1 Revision 2211	加入设备信息接口/System/deviceInfo。	罗展昭	2013-08-30
1.1 Revision 2214	加入设备信息接口 sharpnessLevel，saturationLevel, hueLevel, contrastLevel	罗展昭	2013-09-07
1.2 Revision 2404	加入设备信息接口 flip、mirror	卢冠霖	2013-11-08
1.2 Revision 2442	加入设备信息接口 /system/time/timezome	卢冠霖	2013-11-03
1.3 Revision 2533	加入设备信息接口/SDcard	罗展昭	2013-12-18
1.3 Revision 2549	加入设备信息接口/System/time/ntp。	卢冠霖	2014-1-6
1.3 Revision 2610	加入设备信息接口/Network/Interface	卢冠霖	2014-1-10
1.4 Revision 2810	加入设备信息接口/Image,用于设置图像处理高级属性	卢冠霖	2014-6-24

1.概述

2.约定和缩写

3.传输机制

4.设备搜索

5

5.1 System

5.1.1 Device Information

/NetSDK/System/deviceInfo	
GET	
Description	用于获取设备信息。
Query	None.
Content	None.
Success Return	DeviceInfo JSON Form.
PUT	
Description	用于设置设备信息。
Query	None.
Content	DeviceInfo JSON Form.
Success Return	Response JSON Form
Notes:	

5.1.2 System Local Time

/NetSDK/System/time/localTime	
GET	
Description	用于获取系统本地时间信息。

Query	None.
Content	None.
Success Return	ISO8601 Date-Time Format String.
PUT	
Description	用于更新系统本地时间信息。
Query	None.
Content	ISO8601 Date-Time Format String.
Success Return	Response JSON Form
Notes: 如北京时间 2013 年 8 月 16 日下午 5 点 30 分 8 秒, 可以表示为 2004-05-03T17:30:08+08:00。	

5.1.3 System Time Ntp

/NetSDK/System/time/ntp[/properties]	
GET	
Description	用于请求同步国际标准时间。
Query	None.
Content	None.
Success Return	TimeNtp JSON Form
PUT	
Description	用于设置 ntp 配置信息。
Query	None.
Content	TimeNtp JSON Form
Success Return	Response JSON Form
Notes:	

5.2 Audio

5.2.1 Audio Input Channel List

/NetSDK/Audio/input/channels[/properties]	
GET	
Description	用于获取所有音频输入通道配置。
Query	None.
Content	None.
Success Return	AudioInputChannel Array JSON Form.
Notes:	

5.2.2 Audio Input Channel

/NetSDK/Audio/input/channel/ID[/properties]

GET	
Description	用于获取指定音频输入通道配置。
Query	None.
Content	None.
Success Return	AudioInputChannel JSON Form.
PUT	
Description	用于更新指定音频输入通道配置。
Query	None.
Content	AudioInputChannel JSON Form.
Success Return	Response JSON Form.
Notes:	

5.2.3 Audio Encode Channel List

/NetSDK/Audio/encode/channels[/properties]	
GET	
Description	用于获取音频编码通道配置列表。
Query	None.
Content	None.
Success Return	AudioEncodeChannel Array JSON Form.
Notes:	

5.2.4 Audio Encode Channel

/NetSDK/Audio/encode/channel/ID[/properties]	
GET	
Description	用于获取指定音频编码通道配置。
Query	None.
Content	None.
Success Return	AudioEncodeChannel JSON Form.
PUT/POST	
Description	用于更新指定音频编码通道配置。
Query	None.
Content	AudioEncodeChannel JSON Form.
Success Return	Response JSON Form.
Notes:	

5.3 Video

5.3.1 Video Input Channel List

/NetSDK/Video/input/channels[/properties]	
GET	
Description	用于获取视频输入通道配置列表。
Query	None.
Content	None.
Success Return	VideoInputChannel Array JSON Form.
Notes:	

5.3.2 Video Input Channel

/NetSDK/Video/input/channel/ID[/properties]	
GET	
Description	用于获取视频输入通道配置列表。
Query	None.
Content	None.
Success Return	VideoInputChannel JSON Form.
PUT	
Description	用于更新视频输入通道配置列表。
Query	None.
Content	VideoInputChannel JSON Form.
Success Return	Response JSON Form.
Notes:	

5.3.3 Video Input Channel Sharpness Level

/NetSDK/Video/input/channel/ID/sharpnessLevel	
GET	
Description	用于获取视频输入通道锐度配置值。
Query	None.
Content	None.
Success Return	Interger.
PUT	
Description	用于更新视频输入通道锐度配置值。
Query	None.
Content	Interger.
Success Return	Response JSON Form.
Notes:	

Range from 0 to 100.

5.3.4 Video Input Channel brightness Level

/NetSDK/Video/input/channel/ID/brightnessLevel	
GET	
Description	用于获取视频输入通道亮度配置值。
Query	None.
Content	None.
Success Return	Interger.
PUT	
Description	用于更新视频输入通道亮度配置值。
Query	None.
Content	Interger.
Success Return	Response JSON Form.
Notes:	

5.3.5 Video Input Channel flipEnabled

/NetSDK/Video/input/channel/ID/flip	
GET	
Description	用于获取视频输入通道 flip 配置值。
Query	None.
Content	None.
Success Return	Boolean
PUT	
Description	用于更新视频输入通道 flip 配置值。
Query	None.
Content	Boolean
Success Return	Response JSON Form.
Notes:	

5.3.6 Video Input Channel mirrorEnabled

/NetSDK/Video/input/channel/ID/mirror	
GET	
Description	用于获取视频输入通道 mirror 配置值。
Query	None.
Content	None.
Success Return	Boolean
PUT	
Description	用于更新视频输入通道 mirror 配置值。
Query	None.
Content	Boolean

Success Return	Response JSON Form.
Notes:	

5.3.7 Video Input Channel Saturation Level

/NetSDK/Video/input/channel/ID/saturationLevel	
GET	
Description	用于获取视频输入通道饱和度配置值。
Query	None.
Content	None.
Success Return	Interger.
PUT	
Description	用于更新视频输入通道饱和度配置值。
Query	None.
Content	Interger.
Success Return	Response JSON Form.
Notes:	
Range from 0 to 100.	

5.3.8 Video Input Channel Hue Level

/NetSDK/Video/input/channel/ID/hueLevel	
GET	
Description	用于获取视频输入通道色度配置值。
Query	None.
Content	None.
Success Return	Interger.
PUT	
Description	用于更新视频输入通道色度配置值。
Query	None.
Content	Interger.
Success Return	Response JSON Form.
Notes:	
Range from 0 to 100.	

5.3.9 Video Input Channel Contrast Level

/NetSDK/Video/input/channel/ID/contrastLevel	
GET	
Description	用于获取视频输入通道对比度配置值。
Query	None.
Content	None.
Success Return	Interger.
PUT	
Description	用于更新视频输入通道对比度配置值。

Query	None.
Content	Interger.
Success Return	Response JSON Form.
Notes: Range from 0 to 100.	

5.3.10 Video Input Channel Privacy Mask List

/NetSDK/Video/input/channel/ID/privacyMasks[/properties]	
GET	
Description	用于获取指定视频输入通道隐私区域配置列表。
Query	None.
Content	None.
Success Return	VideoPrivacyMask Array JSON Form.
Notes:	

5.3.11 Video Input Channel Privacy Mask

/NetSDK/Video/input/channel/ID/privacyMask/ID[/properties]	
GET	
Description	用于获取指定视频输入通道上指定隐私区域配置。
Query	None.
Content	None.
Success Return	VideoPrivacyMask JSON Form.
PUT	
Description	用于更新指定视频输入通道上指定隐私区域配置。
Query	None.
Content	VideoPrivacyMask JSON Form.
Success Return	Response JSON Form.
Notes:	

5.3.12 Video Encode Channel List

/NetSDK/Video/encode/channels[/properties]	
GET	
Description	用于获取视频编码通道配置列表。
Query	None.
Content	None.
Success Return	VideoEncodeChannel Array JSON Form.
Notes:	

5.3.13 Video Encode Channel

/NetSDK/Video/encode/channel/ID[/properties]	
GET	
Description	用于获取指定视频编码通道配置。
Query	None.
Content	None.
Success Return	VideoEncodeChannel JSON Form.
PUT	
Description	用于更新指定视频编码通道配置。
Query	None.
Content	VideoEncodeChannel JSON Form.
Success Return	Response JSON Form.
Notes:	
为了支持多个视频输入设备，在 URL 中的视频输入通道 ID 需要包含视频输入通道 ID，因此定义为：视频编码码流 ID+视频输入通道 ID*100，例如，视频输入通道 1 的第一个视频编码码流对应的 URL 应该为/NetSDK/Video/encode/channel/101，第二个视频编码码流对应的 URL 应该为/NetSDK/Video/encode/channel/102。	

5.3.14 Video Encode Channel Name Overlay

/NetSDK/Video/encode/channel/ID/channelNameOverlay[/properties]	
GET	
Description	用于获取指定视频编码通道上通道名称叠加层配置。
Query	None.
Content	None.
Success Return	VideoChanelNameOverlay JSON Form.
PUT	
Description	用于更新指定视频编码通道上通道名称叠加层配置。
Query	None.
Content	VideoChanelNameOverlay JSON Form.
Success Return	Response JSON Form.
Notes:	

5.3.15 Video Encode Datetime Overlay

/NetSDK/Video/encode/channel/ID/datetimeOverlay[/properties]	
GET	
Description	用于获取指定视频编码通道上日期时间叠加层配置。
Query	None.
Content	None.
Success Return	VideoDatetimeOverlay JSON Form.
PUT	
Description	用于更新指定视频编码通道上日期时间叠加层配置。
Query	None.

Content	VideoDatetimeOverlay JSON Form.
Success Return	Response JSON Form.
Notes:	

5.3.16 Video Encode Deivce ID Overlay

/NetSDK/Video/encode/channel/ID/deviceIDOverlays[/properties]	
GET	
Description	用于获取指定视频编码通道上设备序号叠加层配置列表。
Query	None.
Content	None.
Success Return	VideoDeviceIDOverlay JSON Form.
PUT	
Description	用于更新指定视频编码通道上设备序号叠加层配置列表。
Query	None.
Content	VideoDeviceIDOverlay JSON Form.
Success Return	Response JSON Form.
Note:	

5.3.17 Video Encode Text Overlay List

/NetSDK/Video/encode/channel/ID/textOverlays[/properties]	
GET	
Description	用于获取指定视频编码通道上文本叠加层配置列表。
Query	None.
Content	None.
Success Return	VideoTextOverlay Array JSON Form.
Notes:	
该接口保留。	

5.3.18 Video Encode Text Overlay

/NetSDK/Video/encode/channel/ID/textOverlay/ID[/properties]	
GET	
Description	用于获取指定视频编码通道上指定文本叠加层配置。
Query	None.
Content	None.
Success Return	VideoTextOverlay JSON Form.
PUT	
Description	用于更新指定视频编码通道上指定文本叠加层配置。
Query	None.
Content	VideoTextOverlay JSON Form.

Success Return	Response JSON Form.
Notes: 该接口保留。	

5.3.19 Video Encode Request Key Frame

/NetSDK/Video/encode/channel/ID/requestKeyFrame	
PUT	
Description	用于对指定视频编码通道请求一个关键帧。
Query	None.
Content	None.
Success Return	Response JSON Form.
Notes:	

5.3.20 Video Encode Snap Shot

/NetSDK/Video/encode/channel/ID/snapShot		
GET		
Description	用于获取指定视频编码通道快照图片。	
Query	resolutionWidth	integer
	resolutionHeight	integer
Content	None.	
Success Return	JPEG over HTTP。	
Notes:		

5.3.21 Video Motion Detection Channel List

/NetSDK/Video/motionDetection/channels[/properties]	
GET	
Description	用于获取视频运动侦测通道配置列表。
Query	None.
Content	None.
Success Return	VideoMotionDetectionChannel Array JSON Form.
Notes:	

5.3.22 Video Motion Detection Channel

/NetSDK/Video/motionDetection/channel/ID[/properties]	
GET	
Description	用于获取指定视频运动侦测通道配置。

Query	None.
Content	None.
Success Return	VideoMotionDetectionChannel JSON Form.
PUT	
Description	用于更新指定视频运动侦测通道配置。
Query	None.
Content	VideoMotionDetectionChannel JSON Form.
Success Return	Response JSON Form.
Notes:	

5.3.23 Video Motion Detection Channel Status

/NetSDK/Video/motionDetection/channel/ID/status	
GET	
Description	用于获取指定视频运动侦测通道状态。
Query	None.
Content	None.
Success Return	Boolean
DELETE	
Description	用于清除指定视频运动侦测通道状态。
Query	None.
Content	None.
Success Return	Response JSON Form.
Notes:	

5.4 IO

5.4.1 IO Alarm Input Channel List

/NetSDK/IO/alarmInput/channels[/properties]	
GET	
Description	用于获取报警输入通道配置列表。
Query	None.
Content	None.
Success Return	AlarmInputChannel Array JSON Form.
Notes:	

5.4.2 IO Alarm Input Channel

/NetSDK/IO/alarmInput/channel/ID[/properties]	
GET	
Description	用于获取指定报警输入通道配置。
Query	None.
Content	None.
Success Return	AlarmInputChannel JSON Form.
PUT	
Description	用于更新指定报警输入通道配置。
Query	None.
Content	AlarmInputChannel JSON Form.
Success Return	Response JSON Form.
Notes:	

5.4.3 IO Alarm Input Channel Port Status

/NetSDK/IO/alarmInput/channel/ID/portStatus	
GET	
Description	用于获取指定报警输入通道状态。
Query	None.
Content	None.
Success Return	AlarmIOPortStatus JSON Form.
Notes:	

5.4.4 IO Alarm Output Channel List

/NetSDK/IO/alarmOutput/channels[/properties]	
GET	
Description	用于获取报警输出通道配置列表。
Query	None.
Content	None.
Success Return	AlarmOutputChannel Array JSON Form.
Notes:	

5.4.5 IO Alarm Output Channel

/NetSDK/IO/alarmOutput/channel/ID[/properties]	
GET	
Description	用于获取指定报警输出通道配置。
Query	None.
Content	None.

Success Return	AlarmOutputChannel JSON Form.
PUT	
Description	用于更新指定报警输出通道配置。
Query	None.
Content	AlarmOutputChannel JSON Form.
Success Return	Response JSON Form.
Notes:	

5.4.6 IO Alarm Output Channel Port Status

/NetSDK/IO/alarmOutput/channel/ID/portStatus	
GET	
Description	用于获取指定报警输出通道状态。
Query	None.
Content	None.
Success Return	AlarmIOPortStatus JSON Form.
DELETE	
Description	用于清除指定报警输出通道状态。
Query	None.
Content	None.
Success Return	Response JSON Form.
Notes:	

5.4.8 IO Alarm Output Channel Trigger

/NetSDK/IO/alarmOutput/channel/ID/trigger		
PUT		
Description	用于手动触发指定报警输出通道输出。	
Query	outputState	low high pulse
Content	None.	
Success Return	Response JSON Form.	
Notes:		

5.5 PTZ

5.5.1 PTZ Channel List

/NetSDK/PTZ/channels[/properties]	
GET	
Description	用于获取云台通道配置列表。
Query	None.
Content	None.
Success Return	PTZChannel Array JSON Form.
Notes:	

5.5.2 PTZ Channel

/NetSDK/ PTZ/channel/ID[/properties]	
GET	
Description	用于获取指定云台通道配置。
Query	None.
Content	None.
Success Return	PTZChannel JSON Form.
PUT	
Description	用于更新指定云台通道配置。
Query	None.
Content	PTZChannel JSON Form.
Success Return	Response JSON Form.
Notes:	

5.5.3 PTZ Channel Control

/NetSDK/ PTZ/channel/ID/control		
PUT		
Description	用于控制指定云台通道。	
Query	command:string	pan_left pan_right tilt_up tilt_down up_left up_right down_left down_right zoom_in zoom_out focus_near focus_far iris_open

		iris_close auto_pan set_preset clear_preset goto_preset
	mode:string	start stop
	speed:integer	0-100
	presetNo:integer	
Content	None.	
Success Return	Response JSON Form.	
Notes:		

5.6 Network Transport Stream

5.6.1 Network Transport Stream Channel List

/NetSDK/Stream/channels	
GET	
Description	用于获取网络传输码流通道配置列表。
Query	None.
Content	None.
Success Return	StreamChannel Array JSON Form.
Notes:	

5.6.2 Network Transport Stream Channel

/NetSDK/Stream/channel/ID	
GET	
Description	用于获取指定网络传输码流通道配置。
Query	None.
Content	None.
Success Return	StreamChannel JSON Form.
Notes:	

5.7 Network

5.7.1 Network Interface List

/NetSDK/Network/interfaces[/properties]	
GET	
Description	用于获取网络接口配置列表。
Query	None.
Content	None.
Success Return	NetworkInterfaceList JSON Form.
Notes:	

5.7.2 Network Interface

/NetSDK/Network/interfaces/ID[/properties]	
GET	
Description	用于获取指定网络接口配置。
Query	None.
Content	None.
Success Return	NetworkInterfaceList JSON Form.
PUT	
Description	用于更新指定网络接口配置。
Query	None.
Content	NetworkInterfaceList JSON Form.
Success Return	Response JSON Form.
Notes:列表有 4 个网络接口，其中 id 为 1 的是本地 IP 配置，id 为 2 的是保留接口，id 为 3 的是默认 IP 配置，id 为 4 的是 wifi 配置，请留意 interfaceName 属性	

5.7.3 Network Interface Lan

/NetSDK/Network/interfaces/ID/lan[/properties]	
GET	
Description	用于获取指定网络接口 LAN 配置。
Query	None.
Content	None.
Success Return	NetworkInterfaceLan JSON Form.
PUT	
Description	用于更新指定网络接口 LAN 配置。
Query	None.
Content	NetworkInterfaceLan JSON Form.
Success Return	Response JSON Form.
Notes:	

5.7.4 Network Interface Pppoe

/NetSDK/Network/interfaces/ID/pppoe[/properties]	
GET	
Description	用于获取指定网络接口配置。
Query	None.
Content	None.
Success Return	NetworkInterfacePppoe JSON Form.
PUT	
Description	用于更新指定网络接口配置。
Query	None.
Content	NetworkInterfacePppoe JSON Form.
Success Return	Response JSON Form.
Notes:	

5.7.5 Network Interface Dns

/NetSDK/Network/interfaces/ID/ddns[/properties]	
GET	
Description	用于获取指定网络接口 DDNS 配置。
Query	None.
Content	None.
Success Return	NetworkInterfaceDdns JSON Form.
PUT	
Description	用于更新指定网络接口 DDNS 配置。
Query	None.
Content	NetworkInterfaceDdns JSON Form.
Success Return	Response JSON Form.
Notes:	

5.7.6 Network Interface Wireless

NetSDK/Network/interfaces/ID/wireless[/properties]	
GET	
Description	用于获取指定网络接口 WIRELESS 配置。
Query	None.
Content	None.
Success Return	NetworkInterfaceWireless JSON Form.
PUT	
Description	用于更新指定网络接口 WIRELESS 配置。
Query	None.
Content	NetworkInterfaceWireless JSON Form.
Success Return	Response JSON Form.

Notes:

5.7.7 Network Port List

/NetSDK/Network/Ports[/properties]	
GET	
Description	用于获取所有网络端口配置列表。
Query	None.
Content	None.
Success Return	NetworkPortList JSON Form.
Notes: web/data 端口只是用 ID 为 1 的，其余为保留接口	

5.7.8 Network Port

/NetSDK/Network/Port/ID[/properties]	
GET	
Description	用于获取指定网络端口配置。
Query	None.
Content	None.
Success Return	NetworkPortList JSON Form.
PUT/POST	
Description	用于更新指定网络端口配置。
Query	None.
Content	NetworkPortList JSON Form.
Success Return	Response JSON Form.
Notes:	

5.7.9 Network Dns

/NetSDK/Network/Dns[/properties]	
GET	
Description	用于获取网络 DNS 配置。
Query	None.
Content	None.
Success Return	NetworkDns JSON Form.
PUT/POST	
Description	用于更新网络 DNS 配置。
Query	None.
Content	NetworkDns JSON Form.
Success Return	Response JSON Form.
Notes:	

5.7.10 Network Esee

/NetSDK/Network/Esee	
GET	
Description	用于获取 ESEE 配置。
Query	None.
Content	None.
Success Return	NetworkEsee JSON Form.
PUT/POST	
Description	用于更新 ESEE 配置。
Query	None.
Content	NetworkEsee JSON Form.
Success Return	Response JSON Form.
Notes:	

5.8 SDCard

5.8.1 SDCard media search

/NetSDK/SDCard/media/search	
GET	
Description	用于获取 SDCard media
Query	None.
Content	None.
Success Return	SDCardDbMedia JSON Form.
Notes:	
按 beginUTC, endUTC 获取 SDCsrd db media	

5.8.2 SDCard media playbackFLV

/NetSDK/SDCard/media/playbackFLV	
GET	
Description	用于回放 SDCard media
Query	None.
Content	None.
Success Return	Integer 0.
Notes:	

5.8.3 SDCard format

/NetSDK/SDCard/format	
GET	
Description	用于格式化 SDCard
Query	None.

Content	None.
Success Return	Integer 0.
Notes:	

5.8.4 SDCard status

/NetSDK/SDCard/status	
GET	
Description	用于获取 SDCard 状态
Query	None.
Content	None.
Success Return	Sdcard status.
Note: Sdcard status:1、work 2、ejected 3、fserror 4、formatting 5、zombie	

5.9 Image

5.9.1 Image

/NetSDK/Image[/properties]	
GET	
Description	用于获取图像处理配置
Query	None.
Content	None.
Success Return	Image JSON Form.
PUT/POST	
Description	用于设置图像处理配置
Query	None.
Content	Image JSON Form.
Success Return	Response JSON Form.
Notes: 查询用get, 设置用put, 设置成功后返回ok的json信息。	

5.9.2 IrCutFilter

/NetSDK/Image/irCutFilter[/properties]	
GET	
Description	用于获取红外配置
Query	None.
Content	None.
Success Return	IrCutFilter JSON Form.
PUT/POST	

Description	用于设置红外配置
Query	None.
Content	IrCutFilter JSON Form.
Success Return	Response JSON Form.
Notes: 查询用 get, 设置用 put, 设置成功后返回 ok 的 json 信息。	

5.9.3 ManualSharpness

/NetSDK/Image/manualSharpness[/properties]	
GET	
Description	用于获取手动锐度配置
Query	None.
Content	None.
Success Return	ManualSharpness JSON Form.
PUT/POST	
Description	用于设置手动锐度配置
Query	None.
Content	ManualSharpness JSON Form.
Success Return	Response JSON Form.
Notes: 查询用 get, 设置用 put, 设置成功后返回 ok 的 json 信息。	

5.9.4 Denoise3D

/NetSDK/Image/denoise3d[/properties]	
GET	
Description	用于获取 3D 去噪配置
Query	None.
Content	None.
Success Return	Denoise3D JSON Form.
PUT/POST	
Description	用于设置 3D 去噪配置
Query	None.
Content	Denoise3D JSON Form.
Success Return	Response JSON Form.
Notes: 查询用 get, 设置用 put, 设置成功后返回 ok 的 json 信息。	

5.9.5 WDR

/NetSDK/Image/wdr[/properties]	
GET	
Description	用于获取宽动态配置
Query	None.
Content	None.

Success Return	WDR JSON Form.
PUT/POST	
Description	用于设置宽动态配置
Query	None.
Content	WDR JSON Form.
Success Return	Response JSON Form.
Notes: 查询用 get，设置用 put，设置成功后返回 ok 的 json 信息。	

5.9.6 AF

/NetSDK/Image/AF	
GET	
Description	用于获取自动对焦统计
Query	None.
Content	None.
Success Return	Unsigned short: Focus Metrics Value (text/html)
DELETE	
Description	用于清除自动对焦统计推送
Query	None.
Content	None
Success Return	Response JSON Form
Notes: 查询用 get，get 后设备会对该 socket 进行主动推送 HTTP 数据包，推送失败 100 次会自动清除推送状态。主动清除用 delete，清除成功后返回 ok 的 json 信息。	

Appendix I Data Structure

DeviceInfo

```
{
    /* rw, string, */
    "deviceName",
    /* rw, integer, "1-128", */

```

```

    "deviceAddress",
    /* ro, string, */
    "description",
    /* ro, string, */
    "model",
    /* ro, string, */
    "serialNumber",
    /* ro, string, */
    "macAddress",
    /* ro, string, */
    "firmwareVersion",
    /* ro, string, */
    "firmwareReleaseDate",
    /* ro, string, */
    "firmwareVersion",
    /* ro, string, */
    "hardwareVersion",
}

```

TimeNtp

```

{
    /* rw, boolean, [true, false] */
    "ntpEnabled",
    /* rw, string, "max": 64, */
    "ntpServerDomain",
}

```

AudioInputChannel

```

{
    /* ro, integer, */
    "id",
    /* ro, string, "input,output", */
    "workMode",
    /* rw, integer, "8000", */
    "sampleRate",
    /* rw, integer, "16", */
    "sampleBitWidth",
    /* rw, integer, "0-100", */
    "inputVolume",
    /* rw, integer, "0-100", */
    "outputVolume",
}

```

AudioEncodeChannel

```

{
    /* ro, integer, */
    "id",
    /* ro, integer, */
    "audioInputChannelID",
    /* rw, boolean, */
    "enabled",
    /* rw, string, "G.711alaw,G.711ulaw", */
    "codecType",
}

```

VideoPrivacyMask

```

{
    /* ro, integer, */
    "id",
    /* ro, boolean, */
    "enabled",
    /* ro, float, "0-100", the relative percentage to capture size */
    "regionX",
    "regionY",
    "regionWidth",
    "regionHeight",
    /* ro, string, the mask color, RGB888 format, such as red "ff0000" */
    "regionColor",
}

```

VideoInputChannel

```

{
    /* ro, integer, */
    "id",
    /* ro, integer, "50,60", */
    "powerLineFrequencyMode",
    /* ro, integer, the video capture size, */
    "captureWidth",
    "captureWidthProperty",
    /* ro, integer, depends on the <powerLineFrequencyMode>, */
    "captureFrameRate",
    /* ro, integer, "0-100", */
    "brightnessLevel",
    "contrastLevel",
    "sharpnessLevel",
    "saturationLevel",
    "hueLevel",
    /* video privacy mask list, */
    "privacyMask" : [],
}

```

VideoChannelNameOverlay

```

{
    /* rw, boolean, indicates the overlay shows or not, */
    "enabled",
    /* rw, float, "0-100", indicates the location of overlay, */
    /* the relative percentage of video size, */
    "regionX",
    "regionY",
}

```

VideoDatetimeOverlay

```

{
    /* rw, boolean, indicates the overlay shows or not, */
    "enabled",
    "regionX",
    "regionY",
    /* rw, string, */
    /* "YYYY/MM/DD,MM/DD/YYYY,DD/MM/YYYY", */
    /* "YYYY-MM-DD,MM-DD-YYYY,DD-MM-YYYY", */
    /* indicates the date format to display, */
    "dateFormat",
    /* rw, integer, "12,24", indicates the time format to display, */
    "timeFormat",
    /* rw, boolean, */
    "displayWeek",
}

```

VideoDeviceIDOverlay

```

{
    /* rw, boolean, indicates the overlay shows or not, */
    "enabled",
    "regionX",
    "regionY",
}

```

VideoTextOverlay

```

{
    /* rw, boolean, indicates the overlay shows or not, */
    "enabled",
    "regionX",
    "regionY",
    /* rw, string, */
    "message",
}

```

VideoEncodeChannel

```

{
    /* ro, integer, */
    "id",
    /* rw, string, */
    "channelName",
    /* rw, boolean, */
    "enabled",
    /* ro, integer, refers to </NetSDK/Video/input/channel/ID>, */
    "videoInputChannelID",
}

```

```

/* rw, string, "H.264", */
"codecType",
/* rw, string, "baseline,main,high", */
"h264Profile",
/* rw, string, format <width x height>, */
"resolution",
/* rw, boolean, */
"freeResolution",
/* rw, integer, indicates the video size when <freeResolution> true, */
"resolutionWidth",
"resolutionHeight",
/* rw, string, "CBR,VBR", */
"bitRateControlType",
/* rw, integer, */
"constantBitRate",
/* rw, integer, */
"frameRate",
/* rw, integer, */
"keyFrameInterval",
/* rw, string, "JPEG" */
"snapshotImageType",
/* the channel name overlay of this encoding, */
"channelNameOverlay",
/* the datetime overlay of this encoding, */
"datetimeOverlay",
/* the device ID overlay of this encoding, */
"deviceIDOverlay",
/* the text overlay of this encoding, */
"textOverlay" : [],
}

```

VideoMotionDetectionGrid

```

{
/* ro, integer, */
"rowGranularity",
/* ro, integer, */
"columnGranularity ",
/* rw, a boolean <rowGranularity> by <columnGranularity> matrix, */
"granularity",
/* ro, integer, "0-100" */
"sensitivityLevel",
}

```

VideoMotionDetectionRegion

```

{
    /* ro, integer, */
    "id",
    /* rw, boolean, */
    "enabled",
    /* rw, float, "0-100" */
    "regionX",
    "regionY",
    "regionWidth",
    "regionHeight",
    /* ro, integer, "0-100" */
    "sensitivityLevel",
}

```

VideoMotionDetectionChannel

```

{
    /* ro, integer, */
    "id",
    /* rw, boolean, */
    "enabled",
    /* rw, string, "grid,region", */
    "detectionType",
    /* refers to <VideoMotionDetectionGrid>, */
    "detectionGrid",
    /* refers to <VideoMotionDetectionRegion>, */
    "detectionRegion" : [],
}

```

AlarmInputChannel

```

{
    /* ro, integer, */
    "id",
    "active" : {
        /* rw, string, "high,low", */
        "defaultState",
        /* rw, string, "high,low", */
        "activeState",
    }
}

```

AlarmOutputChannel

```

{
    /* ro, integer, */
    "id",
}

```

```

    "active" : {
        /* rw, string, "high,low", */
        "defaultState",
        "activeState",
    },
    /* rw, string, "continuous,low", */
    "powerOnState",
    /* rw, integer, "1000-", in millisecond */
    "pulseDuration",
}

```

AlarmIOPortStatus

```

{
    /* ro, string, "input,output", the type of this alarm IO port */
    "type",
    /* ro, string, "high,low", the current state of this alarm IO port */
    "state",
    /* ro, boolean, */
    "active",
}

```

PTZChannel

```

{
    /* ro, integer, */
    "id",
    /* ro, integer, */
    "serialPortID",

    "videoInputID",
    /* ro, string, "half,full", */
    "duplexMode",
    /* ro, string, "internal,external", */
    "controlType",
    "externalControl" : {
        /* rw, string, */
        "protocol",
        /* rw, integer, */
        "address",
    },
}

```

StreamTransportRTSP

```

{
    /* ro, boolean, */
}

```

```

    "enabled",
    /* ro, boolean, */
    "authorization",
    /* rtsp://<ipAddress>[:<port>]/<absolutePath>, */
    /* ro, string, */
    "absolutePath",
    /* ro, integer, "1-60000", */
    "port",
}

```

StreamTransportRTSPoverHTTP

```

{
    /* ro, boolean, */
    "enabled",
    /* ro, boolean, */
    "authorization",
    /* ro, string, "GET", */
    "requestMethod",
    /* http://<ipAddress>[:<port>]/<absolutePath>, */
    /* ro, string, */
    "absolutePath",
    /* ro, integer, "1-60000", */
    "port",
}

```

StreamTransportFLVoverHTTP

```

{
    /* ro, boolean, */
    "enabled",
    /* ro, boolean, */
    "authorization",
    /* ro, string, "GET", */
    "requestMethod",
    /* http://<ipAddress>[:<port>]/<absolutePath>, */
    /* ro, string, */
    "absolutePath",
    /* ro, integer, "1-60000", */
    "port",
}

```

StreamTransportRTMP

```

{
    /* ro, boolean, */
    "enabled",
}

```

```

    /* ro, boolean, */
    "authorization",
    /* rtmp://<ipAddress>[:<port>]/<absolutePath>, */
    /* ro, string, */
    "absolutePath",
    /* ro, integer, "1-60000", */
    "port",
}

```

StreamChannel

```

{
    /* ro, ineteger, */
    "id",
    /* ro, ineteger, refers to </NetSDK/audio/encode/channel/ID>, */
    "audioEncodeChannelID",
    /* ro, ineteger, refers to </NetSDK/video/encode/channel/ID>, */
    "videoEncodeChannelID",
    /* refers to <StreamTransportRTSP>, */
    "transportRTSP",
    /* refers to <StreamTransportRTSPoverHTTP>, */
    "transportRTSPoverHTTP",
    /* refers to <StreamTransportFLVoverHTTP>, */
    "transportFLVoverHTTP",
    /* refers to <StreamTransportRTMP>, */
    "transportRTMP",
}

```

NetworkInterfaceList

```

{
    /* ro, integer, */
    "id",
    /* ro, string, */
    "interfaceName",
    /* refers to <NetworkInterfaceLan>, */
    "lan",
    /* refers to <NetworkInterfaceUpnp>, */
    "upnp",
    /* refers to <NetworkInterfacePppoe>, */
    "pppoe",
    /* refers to <NetworkInterfaceDdns>, */
    "ddns",
    /* refers to <NetworkInterfaceWireless>, */
    "wireless",
    /* refers to <NetworkInterfaceAccessPointMode>, */
}

```

```

    "accessPointMode",
    /* refers to <NetworkInterfaceDhcpServer>, */
    "dhcpServer",
}

```

NetworkInterfaceLan

```

{
    /* rw, string, */
    "ipVersion",
    /* rw, string, ["static", "dynamic"] */
    "addressingType",
    /* rw, string, */
    "staticIP",
    /* rw, string, */
    "staticNetmask",
    /* rw, string, */
    "staticGateway",
}

```

NetworkInterfaceUpnp

```

{
    /* ro, boolean, */
    "enabled",
}

```

NetworkInterfacePppoe

```

{
    /* ro, boolean, */
    "enabled",
    /* rw, string, "max": 64 */
    "pppoeUserName",
    /* rw, string, "max": 64 */
    "pppoePassword",
}

```

NetworkInterfaceDns

```

{
    /* ro, boolean, */
    "enabled",
    /*rw,string, "DYNDNS","NOIP","3322",
    "CHANGEIP","POPDVR","SKYBEST","DVRTOP"*/
    "ddnsProvider",
    /* rw, string, "max": 64 */
    "ddnsUrl",
}

```

```

/* rw, string, "max": 64 */
"ddnsUserName",
/* rw, string, "max": 64 */
"ddnsPassword",
}

```

NetworkInterfaceWireless

```

{
/* rw, string, ["none","accessPoint","stationMode"] */
"wirelessMode",
/* refers to < stationMode >, */
"stationMode",
}

```

NetworkInterfaceStationMode

```

{
/* ro, string, */
"wirelessStaMode",
/* rw, string, "max": 48 */
"wirelessApBssId",
/* rw, string, "max": 32 */
"wirelessApEssId",
/* rw, string, "min": 8,"max": 63 */
"wirelessApPsk",
}

```

NetworkInterfaceAccessPointMode

```

{
/* ro, string, */
"wirelessBssId",
/* rw, string, "max": 32 */
"wirelessEssId",
/* rw, string, "min": 8,"max": 63 */
"wirelessPsk",
/* rw, string, ["802.11b","802.11g","802.11n","802.11bg","802.11bgn"] */
"wirelessApMode",
/*rw,string, ["Auto","1","2","3","4","5",
"6","7","8","9","10","11","12","13","14"]*/
"wirelessApMode80211nChannel",
/* ro,boolean, */
"wirelessEssIdBroadcastingEnabled",
/* rw, string, ["WPA_PSK","WPA2_PSK"] */
"wirelessWpaMode",
}

```

NetworkInterfaceDhcpServer

```
{  
    /* ro, boolean, */  
    "enabled",  
    /* ro, boolean, */  
    "dhcpAutoSettingEnabled",  
    /* rw, string, "max": 64 */  
    "dhcpIpRange",  
    /* rw, string, "max": 32 */  
    "dhcpIpNumber",  
    /* rw, string, "max": 32 */  
    "dhcpIpDns",  
    /* rw, string, "max": 32 */  
    "dhcpIpGateway",  
}
```

NetworkPortList

```
{  
    /* ro, integer, */  
    "id",  
    /* ro, string, */  
    "portName",  
    /* rw, integer, "min" : 1, "max" 60000 */  
    "value",  
}
```

NetworkDNS

```
{  
    /* ro, string, */  
    "preferredDns",  
    /* ro, string, */  
    "staticAlternateDns",  
}
```

NetworkEsee

```
{  
    /* ro, boolean, */  
    "enabled",  
}
```

SDCardDbMedia

```
{  
    /* rw, integer, */
```

```

    "sessionID"
    /* rw, integer, */
    "channelID".
    /* rw, integer, */
    "beginUTC"
    /* rw, integer, */
    "endUTC"
    /* rw, string, */
    "type"
},

```

Image

```

{
    /* rw, string, [ "auto", "indoor", "outdoor"], */
    "sceneMode"
    /* refers to <ImageIrCutFilter>, */
    "irCutFilter"
    /* refers to < ImageManualSharpness>, */
    "manualSharpness"
    /* refers to <ImageDenoise3d>, */
    "denoise3d"
    /* refers to <ImageWDR>, */
    "WDR"
    /* rw, string, ["auto", "bright", "dark"]*/
    "exposureMode"
    /* rw, string, ["auto", "indoor", "outdoor"]*/
    "awbMode"
    /* rw, string, ["close", "only night", "day-night", "auto"]*/
    "lowlightMode"
}

```

IrCutFilter

```

{
    /* rw, string, ["hardware", "software"] */
    "irCutControlMode"
    /* rw, string, ["auto", "daylight", "night"]*/
    "irCutMode"
}

```

ManualSharpness

```

{
    /* ro, boolean, */
    "enabled"
}

```

```
/* rw, integer, "min" : 0, "max" : 255, "def" : 170*/  
"sharpnessLevel"  
}
```

Denoise3d

```
{  
  /* ro, boolean, */  
  "enabled"  
  /* rw, integer, "min" : 1, "max" : 5, "def" : 1*/  
  "denoise3dStrength"  
}
```

WDR

```
{  
  /* ro, boolean, */  
  "enabled"  
  /* rw, integer, "min" : 1, "max" : 5, "def" : 3 */  
  "WDRStrength"  
}
```